

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	C.Mokshitha Reddy	Reg. No.:	192571054
Section / Batch:		Date:	24/07/2026

Code of Conduct

I _____ C.Mokshitha reddy(192571054) _____
(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student:
_____ Mokshitha.C _____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 - 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

SET 1



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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) ← Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List

Q1 Binary Search 5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {  
    int low = 0, high = n - 1;  
    while(low < high) {  
        int mid = (low + high) / 2;  
        if(arr[mid] == key)  
            return mid;  
        else if(arr[mid] < key)  
            low = mid;  
        else  
            high = mid;  
    }  
    return -1;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.



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QUESTIONS

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Your Fixed Code

```
1 #include <stdio.h>  
2 int binarysearch(int arr[],int n,int key)  
3 {  
4     int low = 0,high = n - 1;  
5     while (low <=high)  
6     {  
7         int mid = (low + high)/2;  
8         if (arr[mid]== key)  
9             return mid;  
10        else if (arr[mid] < key)  
11            low = mid + 1;  
12        else  
13            high = mid - 1;  
14    }  
15    return -1;  
16 }  
17 int main(){  
18     int arr[]={2,4,6,8,10,12,14,16,18,20};  
19     int n=sizeof(arr)/sizeof(arr[0]);  
20     int key;  
21     printf("array:");  
22     for(int i=0;i<n;i++)  
23     {  
24         printf("%d",arr[i]);  
25         printf("\n");
```

Test Input

stdin input (optional)

Output

```
array:2  
4  
6  
8  
10  
12
```



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```
18 int arr[]={2,4,6,8,10,12,14,16,18,20};
19 int n=sizeof(arr)/sizeof(arr[0]);
20 int key;
21 printf("array:");
22 for(int i=0;i<n;i++)
23 {
24     printf("%d",arr[i]);
25     printf("\n");
26 }
27 printf("enter key to search:");
28 scanf("%d",&key);
29 int result = binarysearch(arr,n,key);
30 if (result!=-1)
31     printf("element found at index %d\n",result);
32 else
33     printf("element not found");
34 return 0;
35 }
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly linked list

Q2 Stack 5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 100
3 int stack[MAX];
4 int top = -1;
```

Test Input

stdin input (optional)



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5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Debugging task. Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 100
3 int stack[MAX];
4 int top = -1;
5 int pop()
6 {
7     if (top == -1)
8     {
9         printf("underflow\n");
10        return -1;
11    }
12    return stack[top--];
13 }
14 int main()
15 {
16     stack[++top] = 10;
17     stack[++top] = 20;
18     stack[++top] = 30;
19     printf("%d\n",pop());
20     return 0;
21 }
```

Test Input

stdin input (optional)

Output

30

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Submitted



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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List

Q3 Stack 5 marks

Check the snippet and identify the logical error.

int stack[5];
int top;
void init() {
 top = 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 5
3 int stack[MAX];
4 int top;
5 void init()
6 {
7     top = -1;
```

Test Input

stdin input (optional)

Output

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #define MAX 5
3 int stack[MAX];
4 int top;
5 void init()
6 {
7     top = -1;
8 }
9 int main()
10 {
11     init();
12     if (top == -1)
13         printf("stack initialized successfully\n");
14     else
15         printf("initialization failed\n");
16     return 0;
17 }

```

Submitted

Test Input

stdin input (optional)

Output

stack initialized successfully

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q4 Stack 5 marks

Identify the errors in the following snippet.

```

if(ch == ')') {
    while(stack[top] != ')') {
        postfix[j++] = pop();
    }
    pop();
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2
3 char stack[10] = {'(', '+'};
4 char postfix[20];
5 int top = 1, j = 0;
6
7 char pop() {
8     return stack[top--];
9 }
10
11 int main() {
12     char ch = ')';
13
14     if (ch == ')') {
15         while (stack[top] != '(') {
16             postfix[j++] = pop();
17         }
18         pop(); // Remove '('
19     }
20
21     postfix[j] = '\0';
22     printf("Postfix = %s", postfix);
23
24     return 0;
25 }

```

Test Input

Infix Expression: A+B
Current Character:)

Output

Postfix = +

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Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

```

1 #include <stdio.h>
2
3 char stack[10] = {'(', '+'};
4 char postfix[20];
5 int top = 1, j = 0;
6
7 char pop() {
8     return stack[top--];
9 }
10
11 int main() {
12     char ch = ')';
13
14     if (ch == ')') {
15         while (stack[top] != '(') {
16             postfix[j++] = pop();
17         }
18         pop(); // Remove '('
19     }
20
21     postfix[j] = '\0';
22     printf("Postfix = %s", postfix);
23
24     return 0;
25 }

```

Test Input

Infix Expression: A+B
Current Character:)

Output

Postfix = +

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1

Binary Search

5 marks



Q2

Stack

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Stack - Balancing Symbols

5 marks



Q6

Singly Linked List



Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||  
(ch == ']' && stack[top] == '[')) {  
    pop();  
} else {  
    return 0;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 char stack[10] = {'['};  
4 int top = 0;  
5  
6 void pop() {  
7     top--;  
8 }  
9  
10 int main() {  
11     char ch = '[';  
12  
13     if ((ch == '[' && stack[top] == '[') ||  
14         (ch == ']' && stack[top] == '[')) {  
15  
16         pop();  
17         printf("Balanced Brackets");  
18     } else {  
19         printf("Not Balanced");  
20         return 0;  
21     }  
22  
23     return 0;  
24 }
```

Test Input

Bracket: }
Stack Top: {

Q5

Stack - Balancing Symbols

5 marks



Q6

Singly Linked List

5 marks



Q7

Stack

5 marks



Q8

Linked List

5 marks



Q9

Linked List

5 marks



Q10

Queue

5 marks



10/10 answered

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 char stack[10] = {'['};  
4 int top = 0;  
5  
6 void pop() {  
7     top--;  
8 }  
9  
10 int main() {  
11     char ch = '[';  
12  
13     if ((ch == '[' && stack[top] == '[') ||  
14         (ch == ']' && stack[top] == '[')) {  
15  
16         pop();  
17         printf("Balanced Brackets");  
18     } else {  
19         printf("Not Balanced");  
20         return 0;  
21     }  
22  
23     return 0;  
24 }
```

Test Input

Bracket: }
Stack Top: {

Output

Balanced Brackets

🔍 Pending manual review



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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q6
Singly Linked List

5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
```

Test Input

stdin input (optional)

Output



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Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head, *second, *third, *temp;
11
12     head = (struct Node *)malloc(sizeof(struct Node));
13     second = (struct Node *)malloc(sizeof(struct Node));
14     third = (struct Node *)malloc(sizeof(struct Node));
15
16     head->data = 10;
17     head->next = second;
18
19     second->data = 20;
20     second->next = third;
21
22     third->data = 30;
23     third->next = NULL;
24
25     temp = head;
```

Test Input

stdin input (optional)

Output

10 20 30

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Q7 Stack
5 marks

Q8 Linked List
5 marks

Q9 Linked List
5 marks

Q10 Queue
5 marks

10/10 answered

```

11 struct Node { int data; struct Node *next; };
12 head = (struct Node *)malloc(sizeof(struct Node));
13 second = (struct Node *)malloc(sizeof(struct Node));
14 third = (struct Node *)malloc(sizeof(struct Node));
15
16 head->data = 10;
17 head->next = second;
18
19 second->data = 20;
20 second->next = third;
21
22 third->data = 30;
23 third->next = NULL;
24
25 temp = head;
26
27 while (temp != NULL) {
28     printf("%d ", temp->data);
29     temp = temp->next;
30 }
31
32 return 0;
33 }

```

Submitted

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

Q1 Binary Search
5 marks

Q2 Stack
5 marks

Q3 Stack
5 marks

Q4 Stack
5 marks

Q5 Stack - Balancing Symbols
5 marks

Q6 Singly Linked List

Q7 Stack 5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int pop() {
7     if (top == -1) {

```

Test Input

Stack = {10, 20, 30}
top = 2

Output



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Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Your Fixed Code

```

1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int pop() {
7     if (top == -1) {
8         printf("Underflow\n");
9         return -1;
10    } else {
11        return stack[top--];
12    }
13 }
14
15 int main() {
16     int item = pop();
17
18     if (item != -1)
19         printf("Popped element = %d", item);
20
21     return 0;
22 }

```

Submitted

Test Input

```
Stack = {10, 20, 30}
top = 2
```

Output

```
Popped element = 30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List

Q8 Linked List 5 marks

Identify the errors in the following code

```

while current != NULL:
    current.next = prev
    prev = current
    current = current.next

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8

```

Test Input

```
Linked List: 10 -> 20 -> 30 -> NULL
```

Output

```
Reversed List: 30 20 10
```



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Q4

Stack

5 marks

✓

Q5

Stack – Balancing Symbols

5 marks

✓

Q6

Singly Linked List

5 marks

✓

Q7

Stack

5 marks

✓

Q8

Linked List

5 marks

✓

Q9

Linked List

5 marks

✓

Q10

Queue

5 marks

✓

Your Fixed Code

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head, *second, *third;
11     struct Node *current, *prev, *next;
12
13     head = (struct Node *)malloc(sizeof(struct Node));
14     second = (struct Node *)malloc(sizeof(struct Node));
15     third = (struct Node *)malloc(sizeof(struct Node));
16
17     head->data = 10;
18     head->next = second;
19
20     second->data = 20;
21     second->next = third;
22
23     third->data = 30;
24     third->next = NULL;
25

```

Test Input

Linked List: 10 -> 20 -> 30 -> NULL

Output

Reversed List: 30 20 10

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Q10

Queue

5 marks

✓

10/10 answered

```

23     third->data = 30;
24     third->next = NULL;
25
26     prev = NULL;
27     current = head;
28
29     while (current != NULL) {
30         next = current->next;
31         current->next = prev;
32         prev = current;
33         current = next;
34     }
35
36     head = prev;
37
38     printf("Reversed List: ");
39     current = head;
40     while (current != NULL) {
41         printf("%d ", current->data);
42         current = current->next;
43     }
44
45     return 0;
46 }

```

Submitted



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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List

Q9 Linked List5 marks

What is the error in the below code?
new.next = head
head = new

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head = NULL;
```

Test Input
Data = 10

Output
Inserted Node = 10



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Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head = NULL;
11     struct Node *newNode;
12
13     newNode = (struct Node *)malloc(sizeof(struct Node));
14     newNode->data = 10;
15
16     newNode->next = head;
17     head = newNode;
18
19     printf("Inserted Node = %d", head->data);
20
21     return 0;
22 }
```

Test Input
Data = 10

Output
Inserted Node = 10

Pending manual review

Submitted



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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List

Q10 Queue 5 marks

What is missing in the below pseudocode?

```

DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]
        
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2
3 int Q[5] = {10, 20, 30};
4 int front = 0, rear = 2;
5
6 int dequeue() {
7     if (front == -1) {
        
```

Test Input

```

Queue = {10, 20, 30}
front = 0
rear = 2
        
```

Output



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Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Your Fixed Code

```

1 #include <stdio.h>
2
3 int Q[5] = {10, 20, 30};
4 int front = 0, rear = 2;
5
6 int dequeue() {
7     if (front == -1) {
8         printf("Underflow\n");
9         return -1;
10    }
11
12    int item = Q[front++];
13
14    if (front > rear) {
15        front = rear - 1;
16    }
17
18    return item;
19 }
20
21 int main() {
22     int x = dequeue();
23
24     if (x != -1)
25         printf("Deleted element = %d\n", x);
        
```

Test Input

```

Queue = {10, 20, 30}
front = 0
rear = 2
        
```

Output

```

Deleted element = 10
        
```

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Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

```
11
12 int item = Q[front++];
13
14 if (front > rear) {
15     front = rear = -1;
16 }
17
18 return item;
19 }
20
21 int main() {
22     int x = dequeue();
23
24     if (x != -1)
25         printf("Deleted element = %d", x);
26
27     return 0;
28 }
```

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SET 2



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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue

Q1 Stack 5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int peek() {
7     if (top == -1) {
8         printf("Stack Underflow\n");
9         return -1;
10    }
```

Test Input

```
Stack = {10, 20, 30}
top = 2
```

Output

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Debugging task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int peek() {
7     if (top == -1) {
8         printf("Stack Underflow\n");
9         return -1;
10    }
11    return stack[top];
12 }
13
14 int main() {
15     int item = peek();
16
17     if (item != -1)
18         printf("Top element = %d", item);
19
20     return 0;
21 }

```

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Test Input

Stack = {10, 20, 30}
top = 2

Output

Top element = 30

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2) ← Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q2 Circular Queue 5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```

rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
    printf("Queue Full");
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int front = 0, rear = 2;
7

```

Test Input

Vehicle = 40
front = 0
rear = 2

Output



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Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int front = 0, rear = 2;
7
8 void enqueue(int vehicle) {
9     if ((rear + 1) % MAX == front) {
10         printf("Queue Full\n");
11         return;
12     }
13     rear = (rear + 1) % MAX;
14     queue[rear] = vehicle;
15     printf("Vehicle inserted = %d\n", vehicle);
16 }
17
18 int main() {
19     enqueue(40);
20     return 0;
21 }
```

Test Input

```
Vehicle = 40
front = 0
rear = 2
```

Output

```
Vehicle inserted = 40
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Circular

Q3 Stack 5 marks

Point out the error.
if top == MAX:
Overflow

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = 3;
7
8 void push(int item) {
9     if (top == MAX - 1) {
10         printf("Overflow\n");
```

Test Input

```
top = 3
item = 50
```

Output

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = 3;
7
8 void push(int item) {
9     if (top == MAX - 1) {
10         printf("Overflow\n");
11         return;
12     }
13
14     stack[++top] = item;
15     printf("Inserted element = %d", item);
16 }
17
18 int main() {
19     push(50);
20     return 0;
21 }
```

Test Input

```
top = 3
item = 50
```

Output

```
Inserted element = 50
```

Run

Save

Submit Fix

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2) ← Back

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue

Q4 Stack 5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int pop() {
7     if (top == -1) {
8         printf("Underflow\n");
9         return -1;
10     }

```

Test Input

```
Stack = {10, 20, 30}
top = 2
```

Output

```
Popped element = 30
```

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int pop() {
7     if (top == -1) {
8         printf("Underflow\n");
9         return -1;
10    }
11
12    return stack[top--];
13 }
14
15 int main() {
16     int item = pop();
17
18     if (item != -1)
19         printf("Popped element = %d", item);
20
21     return 0;
22 }

```

Test Input

Stack = {10, 20, 30}
top = 2

Output

Popped element = 30

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D5 PROGRAMMING

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Stack
5 marks

Q5 Doubly Linked List 5 marks

Identify the error in this Doubly Linked List node deletion.

```

void deleteNode(Node* del) {
    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;

```

Test Input

Doubly Linked List: 10 <-> 20
Delete Node: 20



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Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

10/10 answered

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *prev, *next;
7 };
8
9 void deleteNode(struct Node **head, struct Node *del) {
10     if (del == NULL)
11         return;
12
13     if (*head == del)
14         *head = del->next;
15
16     if (del->prev != NULL)
17         del->prev->next = del->next;
18
19     if (del->next != NULL)
20         del->next->prev = del->prev;
21
22     free(del);
23 }
24
25 int main() {
26     struct Node *head, *second;

```

Doubly Linked List: 10 <-> 20
Delete Node: 20

Output

Remaining Node = 10

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10/10 answered

```

24
25 int main() {
26     struct Node *head, *second;
27
28     head = (struct Node *)malloc(sizeof(struct Node));
29     second = (struct Node *)malloc(sizeof(struct Node));
30
31     head->data = 10;
32     head->prev = NULL;
33     head->next = second;
34
35     second->data = 20;
36     second->prev = head;
37     second->next = NULL;
38
39     deleteNode(&head, second);
40
41     printf("Remaining Node = %d", head->data);
42
43     free(head);
44     return 0;
45 }

```

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QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q6 Queue 5 marks

Identify the error in the Queue enqueue logic.
void enqueue(int data) {
if (rear == MAX) return;
queue[rear++] = data;
}

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int rear = 2;
7
8 void enqueue(int data) {
9     if (rear == MAX - 1) {
10         printf("Queue Overflow\n");
11     }
12     queue[rear] = data;
13     rear++;
14 }
```

Test Input

rear = 2
data = 40

Output



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QUESTIONS

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int rear = 2;
7
8 void enqueue(int data) {
9     if (rear == MAX - 1) {
10         printf("Queue Overflow\n");
11         return;
12     }
13     queue[rear] = data;
14     rear++;
15 }
16
17 int main() {
18     enqueue(40);
19     printf("Inserted element = %d", queue[rear]);
20
21     return 0;
22 }
```

Test Input

rear = 2
data = 40

Output

Inserted element = 40

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7 Queue - Deque5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int deque[MAX];  
6 int front = 2;  
7
```

Test Input
front = 2
data = 10

Output



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🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7 Queue - Deque5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int deque[MAX];  
6 int front = 2;  
7  
8 void insertFront(int data) {  
9     if (front == 0)  
10         front = MAX - 1;  
11     else  
12         front = front - 1;  
13  
14     deque[front] = data;  
15 }  
16  
17 int main() {  
18     insertFront(10);  
19     printf("Inserted element = %d", deque[front]);  
20     return 0;  
21 }
```

Test Input
front = 2
data = 10

Output
Inserted element = 10

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2) ← Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue

Q8 Binary Search 5 marks

Identify the bug in this Recursive Binary Search.
int binarySearch(int arr[], int l, int r, int x) {
int mid = l + (r - l) / 2;
if (arr[mid] == x) return mid;
if (arr[mid] > x) return binarySearch(arr, l, mid, x);
return binarySearch(arr, mid, r, x);
}

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int l, int r, int x) {
4     if (l > r)
5         return -1;
```

Test Input
Array: 10 20 30 40 50
Search Element: 40



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Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

10/10 answered

```
5     return -1;
6
7     int mid = l + (r - l) / 2;
8
9     if (arr[mid] == x)
10        return mid;
11
12    if (arr[mid] > x)
13        return binarySearch(arr, l, mid - 1, x);
14
15    return binarySearch(arr, mid + 1, r, x);
16 }
17
18 int main() {
19     int arr[] = {10, 20, 30, 40, 50};
20     int x = 40;
21
22     int result = binarySearch(arr, 0, 4, x);
23
24     if (result != -1)
25         printf("Element found at index %d", result);
26     else
27         printf("Element not found");
28
29     return 0;
30 }
```

Output
Element found at index 3

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue

Q9 Circular Queue5 marks

Identify the bug in this Circular Queue dequeue logic.
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX] = {10, 20, 30};
```

Test Input

```
Queue = {10, 20, 30}
front = 0
rear = 2
```



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Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

10/10 answered

```
5 int queue[MAX] = {10, 20, 30};
6 int front = 0, rear = 2;
7
8 int dequeue() {
9     if (front == -1) {
10         printf("Underflow\n");
11         return -1;
12     }
13
14     int data = queue[front];
15
16     if (front == rear)
17         front = rear = -1;
18     else
19         front = (front + 1) % MAX;
20
21     return data;
22 }
23
24 int main() {
25     int x = dequeue();
26
27     if (x != -1)
28         printf("Deleted element = %d", x);
29
30     return 0;
31 }
```

Output
Deleted element = 10

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)
← Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q10 Stack - Postfix Expression
5 marks

Find the bug in the evaluation of a Postfix expression.

```

while (expression[i] != '\0') {
    if (isdigit(expression[i])) push(expression[i] - '0');
    else {
        int op1 = pop();
        int op2 = pop();
        // perform operation: result = op1 + op2;
        push(result);
    }
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

#include <stdio.h>

```

Test Input
Postfix Expression: 23+



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5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

10/10 answered

Your Fixed Code

```

#include <stdio.h>
int stack[10], top = -1;
void push(int x) {
    stack[++top] = x;
}
int pop() {
    return stack[top--];
}
int main() {
    char expression[] = "23+";
    int i = 0, result;
    while (expression[i] != '\0') {
        if (expression[i] >= '0' && expression[i] <= '9')
            push(expression[i] - '0');
        else {
            int op2 = pop();
            int op1 = pop();
            result = op1 + op2;
            push(result);
        }
    }
    printf("%d", result);
}

```

Test Input
Postfix Expression: 23+

Output
Result = 5

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Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

```
10     return stack[top--];  
11 }  
12  
13 int main() {  
14     char expression[] = "23+";  
15     int i = 0, result;  
16  
17     while (expression[i] != '\0') {  
18         if (expression[i] >= '0' && expression[i] <= '9')  
19             push(expression[i] - '0');  
20         else {  
21             int op2 = pop();  
22             int op1 = pop();  
23             result = op1 + op2;  
24             push(result);  
25         }  
26         i++;  
27     }  
28  
29     printf("Result = %d", pop());  
30  
31     return 0;  
32 }
```

Submitted

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